

Ex.No:1: Simulation of FDMA in MATLAB Date

Aim: To simulate a Frequency Division Multiple Access (FDMA) system using MATLAB and analyze its communication performance in terms of channel bandwidth, data rate, and signal-to-noise ratio (SNR).

Objective 1:

To simulate an FDMA system by allocating **individual frequency channels** to multiple users and determine the **channel bandwidth** assigned to each user.

Output Parameter: Channel Bandwidth (kHz)

Objective 2:

To calculate the **data rate** achieved by each user in the FDMA system based on the allocated channel bandwidth.

Output Parameter: Data Rate (kbps)

Objective 3:

To evaluate the **communication quality** of the FDMA system by calculating the **Signal-to-Noise Ratio (SNR)** for different users.

Output Parameter: Signal-to-Noise Ratio (SNR) (dB)

Procedure

1. Open **MATLAB** and create a new script file.
2. Define the total available bandwidth, number of FDMA users, carrier frequencies, signal power, and noise power.
3. Allocate a unique frequency channel to each user and calculate the **channel bandwidth**.
4. Compute the **data rate** for each user based on the allocated bandwidth and modulation scheme.
5. Calculate the **Signal-to-Noise Ratio (SNR)** using the signal and noise power values.
6. Plot the transmitted signals of all FDMA users using different carrier frequencies.
7. Display the calculated **channel bandwidth, data rate, and SNR** in the MATLAB Command Window.
8. Analyze the results to verify independent channel allocation and evaluate the overall performance of the FDMA system.

Program (Matlab Code)

```

clc; clear; close all;

fs=1000; t=0:1/fs:1;

TBW=600; Users=3;

CBW=TBW/Users;

fc=[100 300 500];

m=sin(2*pi*10*t);

s1=m.*cos(2*pi*fc(1)*t);

s2=m.*cos(2*pi*fc(2)*t);

s3=m.*cos(2*pi*fc(3)*t);

subplot(2,1,1)

plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on

title('FDMA Channel Allocation'),xlabel('Time (s)'),ylabel('Amplitude') subplot(2,1,2)

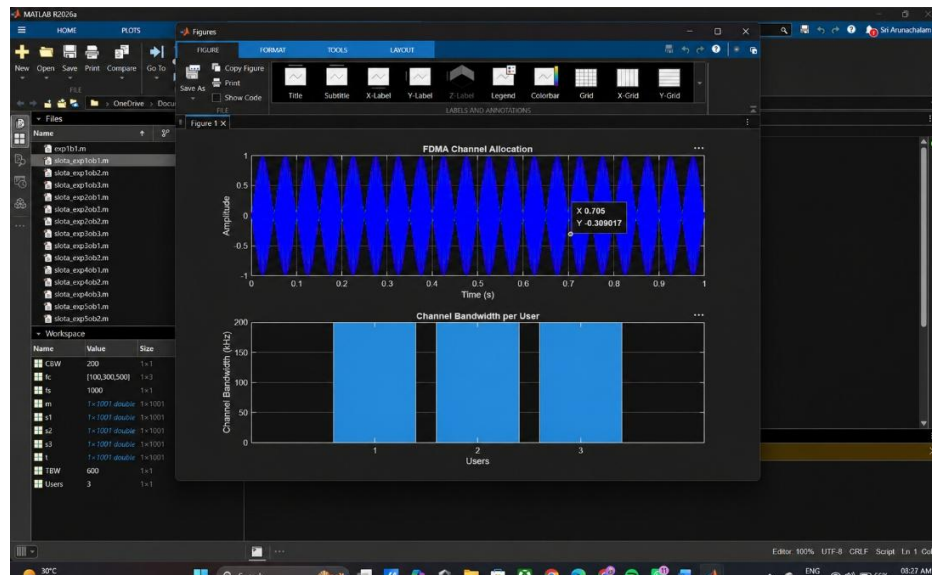
bar(1:Users,CBW*ones(1,Users)),grid on

xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel Bandwidth per User')

```

Output:

Objective 1:



Output Graph 1: FDMA Channel Allocation

- Displays **three users** transmitting on **different carrier frequencies** (100 kHz, 300 kHz, and 500 kHz).
- Demonstrates that each user occupies a separate frequency channel.

Output Graph 2: Channel Bandwidth Allocation

- Shows a **bar graph** of the bandwidth allocated to each user.
- Each user is assigned **200 kHz**, confirming equal FDMA channel allocation.

Objective 2: Matlab Code and Output

```
clc; clear; close all;
```

```
BW = 200; % Channel Bandwidth (kHz)
```

```
Users = 3;
```

```
M = 2; % BPSK Modulation
```

```
DataRate = BW*log2(M); % Data Rate (kbps)
```

```
Rate = DataRate*ones(1,Users);
```

```
disp(' User Data Rate (kbps)')
```

```
disp([(1:Users)' Rate'])
```

```
figure
```

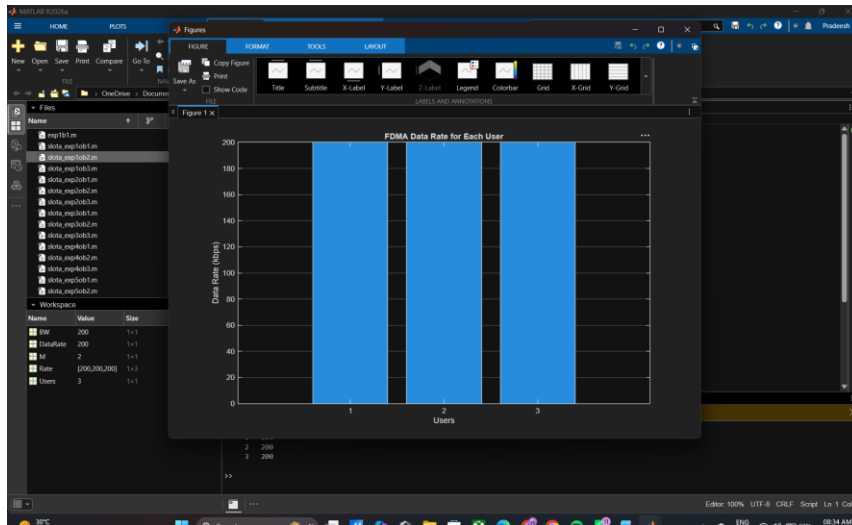
```
bar(1:Users,Rate)
```

```
xlabel('Users')
```

```
ylabel('Data Rate (kbps)')
```

```
title('FDMA Data Rate for Each User')
```

```
grid on
```



The graph shows that all FDMA users achieve an equal **data rate of 200 kbps** due to equal channel bandwidth allocation. This confirms fair and interference-free data transmission among users.

Objective 3: Matlab Code and Ooutput

```
clc; clear; close all;
```

```
Ps = 1; % Signal Power (W)
```

```
Pn = [0.1 0.05 0.01]; % Noise Power (W)
```

```
Users = 1:3;
```

```
SNR = 10*log10(Ps./Pn); % SNR in dB
```

```
disp('User Noise Power(W) SNR(dB)')
```

```
disp([Users' Pn' SNR'])
```

```
figure
```

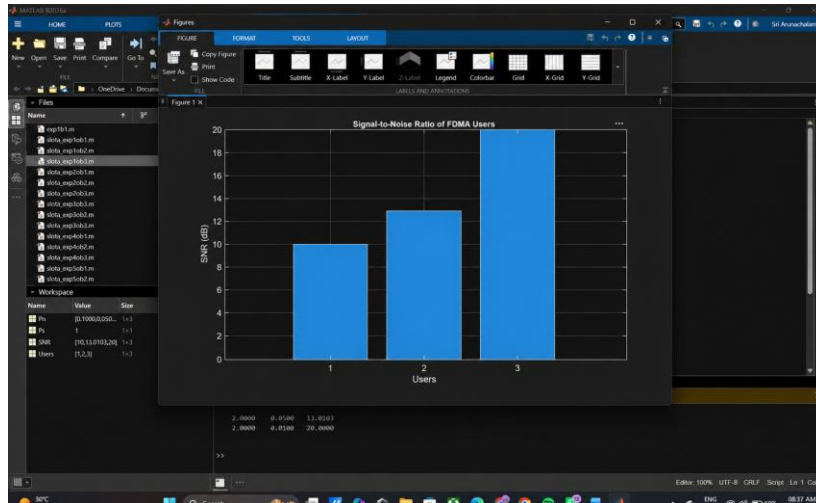
```
bar(Users,SNR)
```

```
xlabel('Users')
```

```
ylabel('SNR (dB)')
```

```
title('Signal-to-Noise Ratio of FDMA Users')
```

```
grid on
```



The graph shows that the **SNR increases as the noise power decreases** for different FDMA users. A higher SNR indicates **better communication quality and more reliable signal transmission**.

Result:

1. The FDMA system was successfully simulated by allocating separate frequency channels to three users, and the **channel bandwidth assigned to each user was determined**.
2. The **data rate achieved by each user** was successfully calculated based on the allocated channel bandwidth, demonstrating equal transmission capacity for all users.
3. The **Signal-to-Noise Ratio (SNR)** for different users was successfully evaluated, showing that higher SNR values provide better communication quality and reliable data transmission.